

PEDRO TAPIA ZAMORA

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EDUCATION

Boston University

Bachelor of Science, Mechanical Engineering

Boston, MA

May 2028

- GPA: 3.51/4.00 (Dean's List)
- Concentration in Energy Technologies & Sustainability, Minor in Mathematics

PROFESSIONAL EXPERIENCE

Public Service Intern | Nevada Department of Transportation

Las Vegas, NV | Summer 2026

- Executed over 20 QA/QC materials tests on aggregates, concrete, and rebar tensile strength.
- Validated structural material compliance through moisture-density testing and OSHA adherence.
- Approved 15+ field and lab examinations in support of a 20-mile highway rehabilitation project with pavement and drainage systems.

Research Assistant | Heejin Cho's Lab at UNLV

Las Vegas, NV | Summer 2026

- Modeled Sun Monitors that track the efficiency of Heliostat Solar Panels in collaboration with Professor Sahm.
- Calculated and applied precision offset algorithms to raw sensor datasets, successfully correcting for temperature-induced drift and wind across 400 points of data.
- Collaborated with PHD students to design a circuit for a Hybrid Battery-Supercapacitor HVAC Emulator.

PROJECTS

Mancherén Grande Water System Design, Boston University Global Engineering Brigade

Summer 2026

- Analyzed water viscosities, pressures, and speeds to achieve optimal flow rate of 5 GPM through Epanet.
- Geologically mapped 100+ acres of land to identify piping placement and elevation changes.
- Surveyed 80 residents to identify water issues and perform water quality assessments.

Hybrid Intro Project, Boston University Rocket Propulsion Group

Winter 2025

- Designed a bell nozzle with an optimized 60% bell contour capable of reaching an altitude of 10,000 feet.
- Performed Finite Element Analysis to model strain and deformation under a 8 lbf force using SOLIDWORKS.
- Manufactured a nozzle via FDM 3D printing using ABS Resin to ensure sufficient strength.

SKILLS

Mechanical: Drill Press, Laser Cutting, Water-Jet Cutting, 3D Printing, Soldering

Software: MATLAB, Python, Microsoft Office, CAD: SOLIDWORKS & Onshape, Epanet, RecTrac, Revit

Relevant Coursework: Statics, Electrical Circuits, Statistics & Probability, and Computational Programming

OTHER EXPERIENCE

Resident Assistant | Boston University Residence Life

Boston, MA | Aug 2025 - Present

- Oversaw 600+ residents during on-call shifts; managed crises with discretion.
- Arranged 20+ educational and social events to promote physical and mental wellness.
- Supervised 55 residents and assisted their transition into college by acting as a mentor figure.

Recreation Assistant & Lead Captain | Walnut Recreation Center

Las Vegas, NV | Jun 2022 - Aug 2026

- Managed front desk operations, coordinated facility reservations and class scheduling to support daily recreation programs.
- Structured and supervised daily activities for children ages 5–12, fostered skill development & positive behavior.
- Streamlined a system to organize 150+ participants in Excel.

LEADERSHIP & AFFILIATIONS

Rocket Propulsion Group, Fluids & Structures Member

Sept 2025 - Present

- Participated in the construction of the Argo Rocket by assisting with the manufacturing of materials.

Global Engineering Brigade, President & Past Director of Fundraising

Sept 2024 - Present

- Recruited new members and went to underdeveloped communities in South America to help with water crises.

Nevada Boys State, Judicial Coordinator & Senior Counselor

Jul 2023 - Present

- Facilitated an environment where High School students can learn about Nevada's legislature and government.